HOMEWORK 9

1. Determine the limit and colimit of the diagram

$$\begin{tikzcd} X \arrow[r, bend left] \arrow[r, bend right] & Y \end{tikzcd}$$

in *Sets*.

2. Prove that all limits and colimits exist in *Sets* and *Groups*.

3. Let $\mathcal{I}$ be a small category with an initial object i and let $F : \mathcal{I} \to \mathcal{C}$ be a functor. Prove that $\lim F = F(i)$.

4. Let $\mathcal{I}$ be a small category and $\mathcal{C}$ a category with limits and colimits. Prove that the functor

$$\mathcal{C} \to Functors(\mathcal{I}, \mathcal{C}),$$

taking an object X to the constant functor c_X, has a left and right adjoint functors. Determine these functors.

5. Prove that a functor that admits a left adjoint functor commutes with limits.

6. Prove that every representable functor $\mathcal{C} \to Sets$ commutes with limits.

7. Let I be a small category and let X be an object of a category $\mathcal{C}$. Prove that the functor $Functors(I, \mathcal{C}) \to Sets$ taking a functor $F : I \to \mathcal{C}$ to $Mor_{\mathcal{C}}(X, \lim F)$ is represented.

8. Prove that the product of two additive categories is additive.

9. Prove that an object X in an additive category is a zero object if and only if $0_X = 1_X$.

10. Let $\mathcal{C}$ be an additive category. Prove that the category of presheaves of abelian groups on $\mathcal{C}$ is also additive.